Marine Microplastic Pollution

Luxi Feyereisen
https://github.com/lulu12320/FWE_CourseWork.git

Introduction

Since the Industrial Revolution, we have been facing a self-destructive environmental issue known as pollution. Pollution comes in many forms, including sound, light, plastic, oil, e-waste, radioactive waste. Among these, plastic waste is one of the most pervasive and persistent. Microplastics are defined as plastic particles that are less than five millimeters. They originate from the breakdown of larger plastic items like bottles, bags, clothes, textiles, and cosmetics. The microplastics soon make their way to the ocean, where they pose a serious threat to marine life. Once ingested by organisms, these toxic microplastics can cause detrimental effects on animal development, reproduction, and resistance to disease (NOAA 2024). As smaller species consume microplastics and are eaten by larger predators, microplastics move throughout the food web by a process called biomagnification. This occurs when higher trophic level species bioaccumulate larger amounts of microplastic toxins. This phenomenon can impact human health and well being, given the global reliance on seafood. This study analyzes the spatial and temporal patterns in microplastic density in the open Pacific Ocean between 2000 and 2010 using data science techniques to explore trends, classify concentration levels, and evaluate predictive modeling approaches.

Methods

This project used data from the National Oceanic and Atmospheric Administration (NOAA), specifically from the National Centers for Environmental Information (NCEI). This dataset contains microplastic concentration measurements from a plethora of sites around the world across several decades. Useful variables such as date, latitude, longitude, microplastic measurement, and concentration class will be analyzed for patterns and trends. This study specifically focused on the open-ocean samples collected by research cruises from the Pacific Ocean between 2000 to 2010. The data was filtered on the NCEI GIS-based Microplastic Map website and downloaded as a CSV file. After importing the data into a Jupyter Notebook, it was cleaned by removing rows with missing or zero values in the microplastic measurement column. Dates were parsed into proper datetime objects while a new column was made to extract the sampling year. The remaining data was then sorted chronologically by date for temporal analysis.

To examine how microplastic pollution varied over time, the dataset was grouped by year and a line plot was created comparing the average annual microplastic density (pieces/m^3) over 10 years (2000 to 2010). To identify geographic microplastic hotspots, a K-Means clustering was applied to the geographic coordinates (latitude and longitude) of each sampling location. The coordinate data was standardized using z-scores before clustering, and each sample was assigned to one of four clusters (k = 4) that reflected a major microplastic hotspot. A scatterplot was made to visualize the spatial distribution of these clusters. This method helps to identify geographic areas with potentially shared environmental characteristics or pollution sources.

K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) classification models were used to evaluate whether location and time could predict pollution levels. These machine learning models used year, latitude, and longitude to predict the concentration class of microplastics. The KNN and SVM model were used as a multiclass classification model with four possible classes (low, medium, high, and very high). The dataset was split into training (70%) and testing (30%) sets to evaluate KNN (k = 3) and SVM (RBF kernel) classifiers. Each model's precision, recall, F1-score, and accuracy was measured using a classification report.

A binary version of the SVM model was used to generate a Receiver Operating Characteristic (ROC) curve and calculate the Area Under the Curve (AUC) score to evaluate its predictive performance between low and medium classes. Again, the dataset was split into 70% training and 30% testing subsets, and recovered a classification report on precision, recall, F1-score, and accuracy metrics. To identify if there was a difference in measured microplastic concentration classes, a two-sample t-test was performed using the difference in means between the low and medium concentration classes. The null hypothesis states that the mean microplastic concentration of the medium and low classes are equal. While the alternative hypothesis states that the mean microplastic concentrations of the medium and low classes are not equal. The test was conducted at a significance level of 0.05.

Results

The temporal line plot revealed a slow upward trend in microplastic density from 2000 to 2009, with a sharp peak in 2009 before dropping back down in 2010 (Figure 1). Spatial analysis using K-Means clustering identified four distinct geographic microplastic hotspots according to sampling coordinates (Figure 2). The KNN classification model achieved strong performance for low, medium, and high classes, but struggled on the underrepresented very high class, with a weighted average precision and recall of 0.76 and an overall accuracy of 76%. On the other hand, the SVM model struggled in identifying most class categories with a 26% precision and 51% accuracy. It especially failed to predict samples from the low, high and very high concentration classes due to the imbalance and insufficient training samples (Figure 3).

The binary SVM classification model was evaluated for its ability to distinguish between low and medium classes. It produced a ROC curve with an AUC score of 0.69 (Figure 4). The two-sample t-test that compared the low and medium concentration classes used a T-statistic of 13.26 and found a p-value of 1.07e-34.

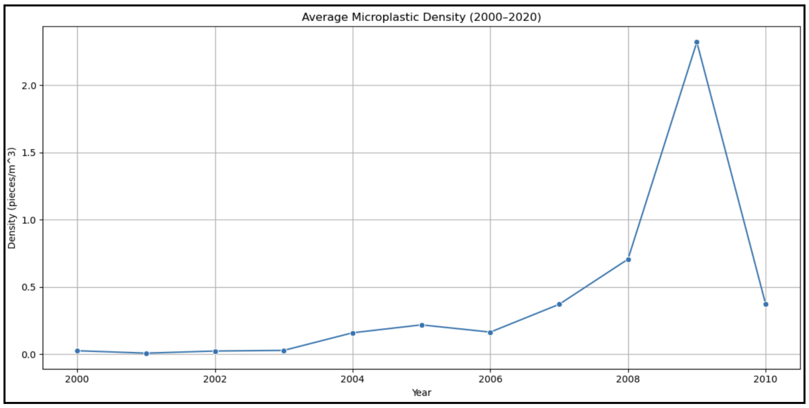

Figure 1: Open Pacific Ocean microplastic density (pieces/m^3) from 2000 to 2010.

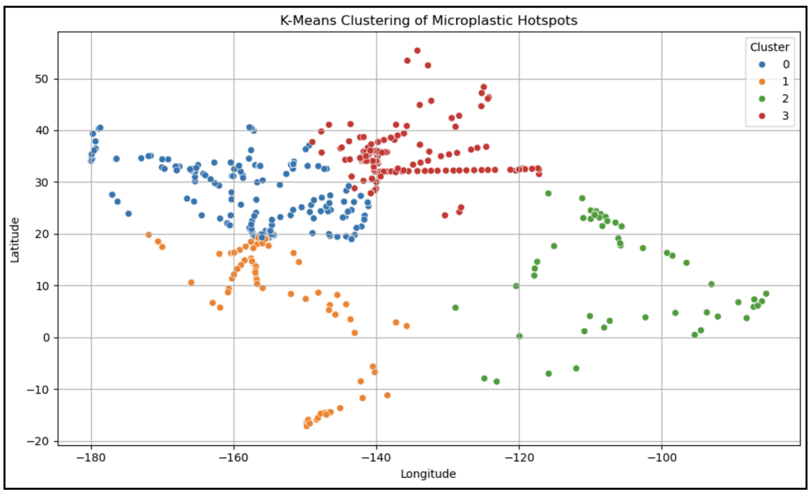

Figure 2: K-Means clustering of microplastic concentration hotspots in the open Pacific Ocean from 2000 to 2010.

```
KNN Classification Report:
              precision    recall  f1-score   support

        High       0.58      0.79      0.67        43
         Low       0.81      0.78      0.79        94
      Medium       0.81      0.76      0.78       148
   Very High       0.00      0.00      0.00         3

    accuracy                           0.76       288
   macro avg       0.55      0.58      0.56       288
weighted avg       0.76      0.76      0.76       288

SVM Classification Report:
              precision    recall  f1-score   support

        High       0.00      0.00      0.00        43
         Low       0.00      0.00      0.00        94
      Medium       0.51      1.00      0.68       148
   Very High       0.00      0.00      0.00         3

    accuracy                           0.51       288
   macro avg       0.13      0.25      0.17       288
weighted avg       0.26      0.51      0.35       288
```

Figure 3: Classification performance of KNN and SVM models on microplastic concentration classes.

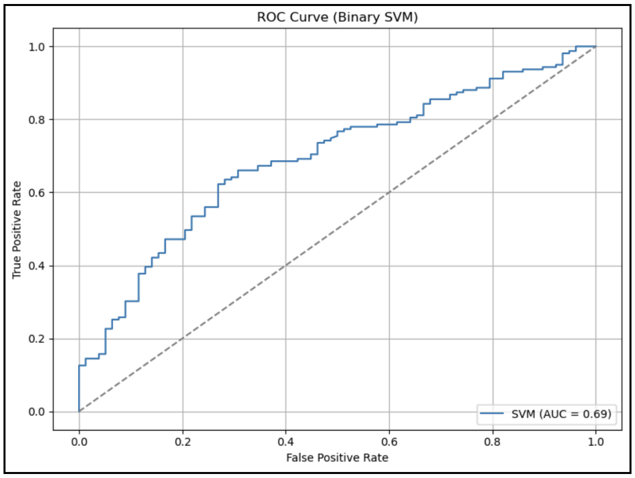

Figure 4: ROC curve for SVM binary classifier predicting microplastic low and medium concentration classes.

Discussion

The results confirmed that microplastic pollution in the open ocean has shown to increase over time but displays some temporal variation with the peak concentration in 2009. The clustering analysis successfully highlights the microplastic hotspots, which are likely associated with ocean currents, human activities, and proximity to urbanization near the coast. While clustering was based only on geographic location, these clusters provide a starting point for spatially targeted investigation. The KNN outperformed the SVM model in classifying concentration classes, suggesting that it is better suited for variable environmental data with an imbalance in class distributions. The SVM model poorly predicted each of the concentration classes, especially low, high, and very high. This suggests that there are limitations of complex models when the data is not uniformly distributed as there were only three samples for the very high concentration class. The binary SVM AUC score of 0.69 indicates moderate predictive reliability. This suggests that year and location data contain some predictive attributes but may be insufficient alone. With the help of additional features like depth and urban proximity, the SVM model could be enhanced and used effectively. The two-sample t-test produced a p-value (1.07e-34) that was much lower than our significance level ($\alpha = 0.05$), thus we can reject the null hypothesis. There was statistically significant evidence that there is a difference in the mean microplastic concentration between medium and low classes.

For future research, predictive performance could be improved by integrating additional features such as ocean current velocity, direction, temperature, sampling depth, microplastic material, and distance to coastlines/urbanization. Expanding the dataset to better represent under-sampled classes like very high would also strengthen machine learning performance. Incorporating recent years of data beyond 2010 could also improve the relevance of model predictions and support real-time monitoring or cleanup planning. The significance in difference of means between low and medium classes confirms that NOAA's classification system does reflect real differences in microplastic density. Overall, the use of machine learning and other data science techniques can help identify present-day high-risk regions and prioritize microplastic mitigation strategies in coastal management and policy development.

References

Serranti, S., Palmieri, R., Bonifazi, G., & Cózar, A. (2018). Characterization of microplastic litter from oceans by an innovative approach based on hyperspectral imaging. Waste Management, 76, 117-125.

US Department of Commerce, National Oceanic and Atmospheric Administration. (2024). NOAA National Ocean Service Education: Coastal Pollution Tutorial. NOAA's National Ocean Service. oceanservice.noaa.gov/education/tutorial-coastal/marine-debris/md04-sub-01.html#:~:text=Lab%20studies%20have%20shown%20that,throughout%20the%20ocean%20food%20chain.